

Project title: from task desc.

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1 TODO

- double col 6-8p.
- technical report, less backround more refs.
- content:

what was done? how was it done? how to reproduce? why even, what's notable (e.g. compared to our existing measurements) results: some proof of work, no you don't need to calculate hashes on paper and hope for zeros - goal: - (me) a broader dataset usable for cycle model improvements based on workloads comparable to real world deployments - as question: How does the performance gap of acae translate to real world representative workloads - plots: - each - future work: - maybe we can even provide some guidelines to performance measurements to improve acae - explicitly which configurations represent proper measurements and which to avoid

Wenn dein Ziel also z.B. ist, die Relevanz von Embench im Vergleich mit Coremark festzustellen, wären Differenzen zwischen den beiden Benchmarks spannend. Oder wenn ich mit Embench z.B. den zeitlichen Ablauf besser erforschen kann, dann wäre ein Plot für die Zeit gut. Usw....

2 STACK

SECTIONS:

- Intro / Motivation / bit of Background / our work
- Results

Motivation:

ACAE currently deviates from real HW performance as proven by (ref ACEA) using Dhrystone and CoreMark In order to improve ACEA to be useful for accurate performance measurements for e.g. rapid prototyping or scientific research both omitting the use of HW, a broader view on comparable performance data is needed. This data should be measured using close to real world workloads, to ensure that measured loads are represented properly by our cycle model. Embench! is a suite provides broader workloads is explicitly built for embedded systems built on BEEBs aims to provide real world workloads runs on hot caches to represent real world

workloads better -> (WILD CLAIM - SOURCE?) e.g. SHA256 performance is comparable to real deployments

Work:

using ACAE in its state as of presented in (ref ACEA) + slight compatibility changes (WHICH? - check git log + diff) we integrated embench as a benchmark suite

CHANGES: acae:

- adjusted Make-System to accomodate benchmark suites by adding the "list" target providing a list of all benchmarks a suite offers embench:
- discarded the python harness -> acae provides its own build system

3 Introduction

Something profound goes here.